Trie/Prefix Tree Review

Do you ever wonder how those autocomplete text options form when you are texting? Or how to store a dictionary?

One could just store each word individually in a massive array. However, really quickly, you'll see that it scales massively and retrieval is horrible $O(n)$. You wouldn't want the program to buffer for absurd amounts of time. What we need is a data structure that gives us $O(\log(n))$ lookup time. The answer lies in the Trie Tree.

What is a Trie Tree?

FORMAL DEFINATION: A Trie is an ordered tree data structure that is used to store a dynamic set or associative array.

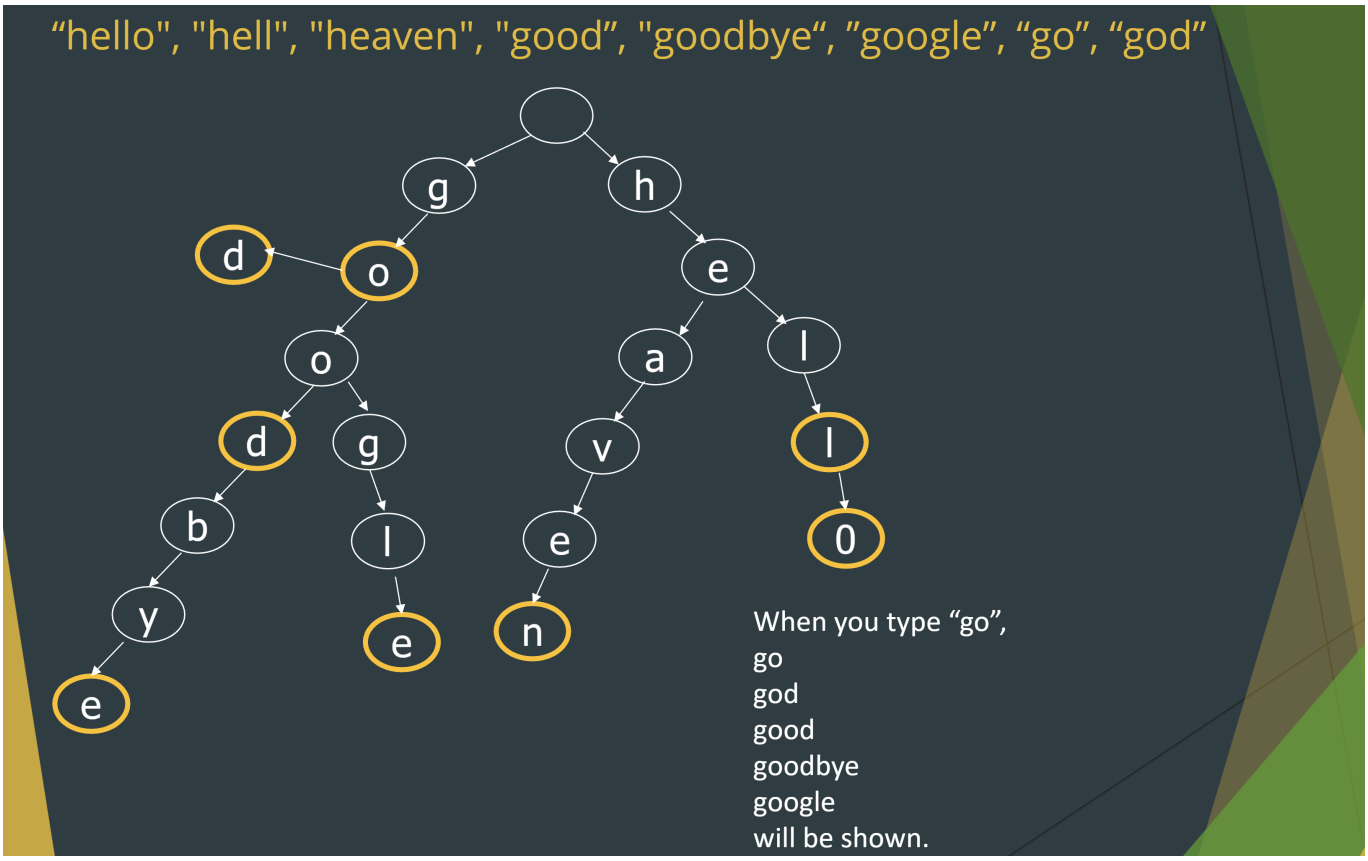
DEFINATION: A trie tree is an ordered tree that can have any amount of children nodes. This means no direction such as left or right node.

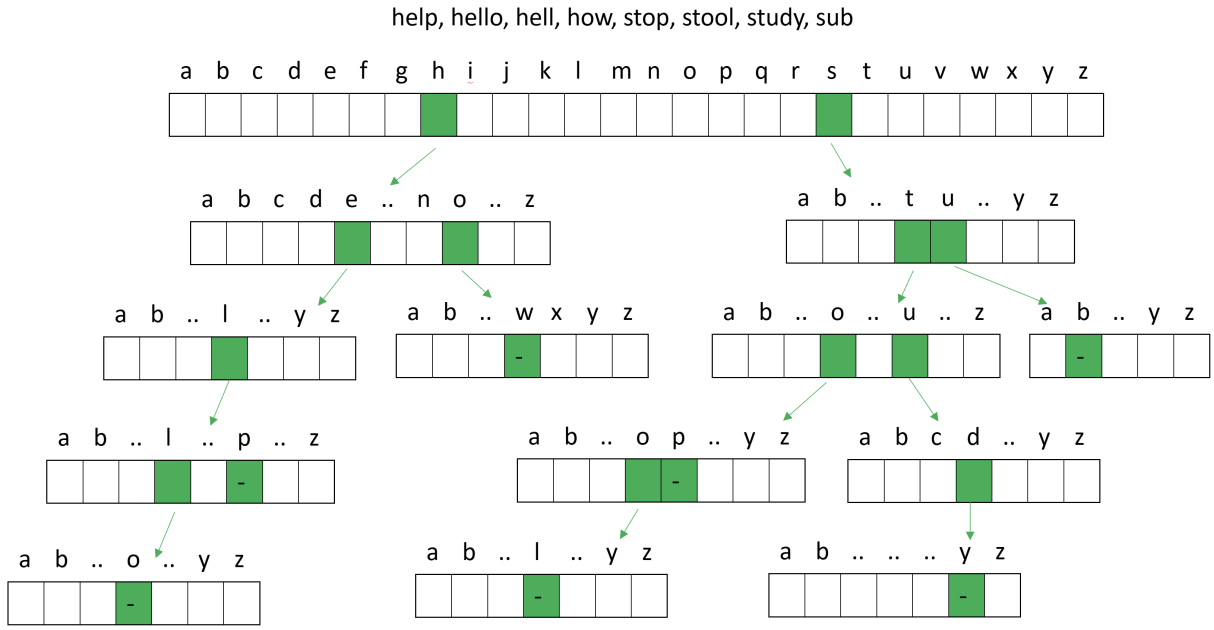
Usages

- Fixed dictionary and want to look up quickly.
- Require less storage for a large dictionary.
- Matching sentences during string matching.

Process for dictionaries

The root will always be an empty because you need it to link to 26 characters of the alphabet for the first character of a word. After that, for each children node, they will each have 26 possible character options and have more children nodes for each added character of a word.





> In the second photo, you can see all of 26 possible nodes.

Code Implementation

```
class tNode{
    public boolean isEnd;
    public tNode[] child = new tNode[26]; // for 26 chars in the alphabet

    public static void addWord(tNode root, String word){
        for (int j = 0; j < word.length(); j++) { // for each letter of the word
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            char c = word.charAt(j); // character at each index

            if(root.child[c-'a']!=null) {  // c-'a' makes it so child[0] is a and
child[25] is z.
                if(word.length()-1 == j) {
                    root.child[c-'a'].isEnd = true;
                }
                root = root.child[c-'a'];
            } else {
                tNode trie = new tNode(); //add child node and iteratively repeat
the process, making the child the new "root".
                trie.isEnd = (word.length()-1 == j ? true:false);
                root.child[c-'a'] = trie; //add the character for the child node.
                root = root.child[c-'a'];
            }
        }
    }

    public static boolean searchWord(tNode root, String word) {
        boolean isFound = true;
        for (int i = 0; i < word.length(); i++) {
            char c = word.charAt(i);

            if (root.child[c - 'a'] != null) { //check through the tree if there
exists a child character node.
                if (word.length() - 1 != i) { // if isn't the last character of
the word, traverse further.
                    root = root.child[c - 'a'];
                } else if (!root.child[c - 'a'].isEnd) { // if in trie tree, it
stops before the entire word is traversed/lookedup.
                    isFound = false;
                }
                // if nothing goes wrong then yay.
            } else { // edge case if even the character exists.
                isFound = false;
                break;
            }
        }
        return isFound;
    }

    public static void printWord(tNode root, String toPrint) {
        if(root==null){ // edge case to check if letter exists in the trie
            return;
        }

        if(root.isEnd){
            System.out.println(toPrint);
        }

        for (int i = 0; i < root.child.length; i++) { // we are printing ALL
POSSIBLE PATHS. root.child.length is always 26.
            tNode t = root.child[i];
```

```
            if(t!=null){ // pass through next node and continually print.
                printWord(t, toPrint + (char)('a' + i )); // recursively add one
character at a time until we reach the isEnd.
            }
        }
    }

    public static void prefixWord(tNode root, String preFix) { // basically
printword but if i want to start with a prefix.
        tNode t= new tNode();
        t=root;
        System.out.println("prefix: " + preFix);

        for (int i = 0; i < preFix.length(); i++) { // basically traverse to the
position in the trie at the end of the prefix. so ignore the past and only the
prefix subtree only.
            char c = preFix.charAt(i);
            t = t.child[c-'a']; // arrive at the spot.
        }
        printWord(t, preFix);
    }

}
```

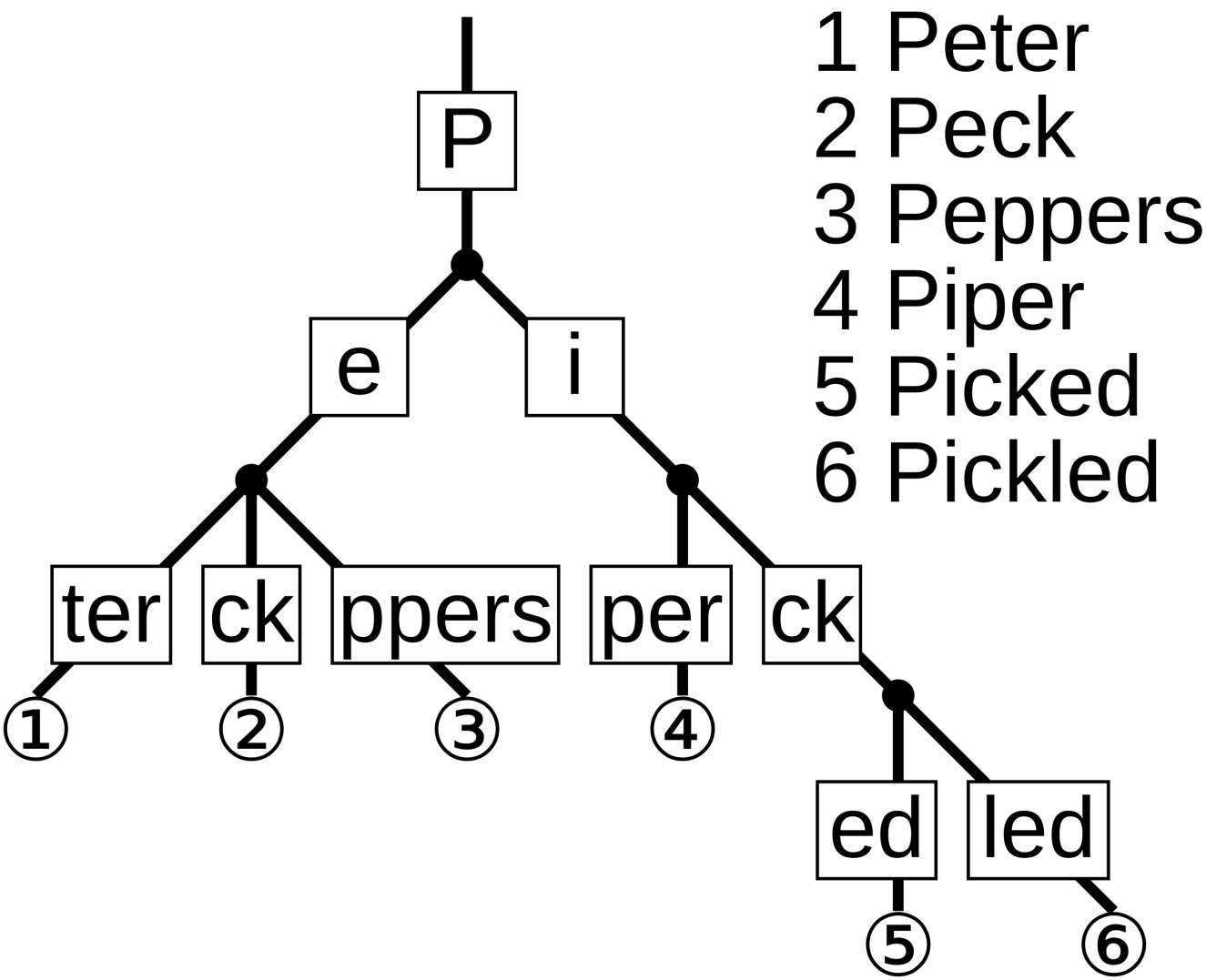
```
public class Trie211 {

    public static void main(String[] args) {
        tNode root = new tNode();
        // add words
        addWord(root, "help");

        // print all the words
        System.out.println("List of words");
        printWord(root, ""); // every single word since we start at empty root.
        System.out.println();

        //ask user for a prefix
        String prefix= "hel";
        prefixWord(root, prefix); // print every word starting with "hel".

        // test if the word is in the tree
        System.out.println(searchWord(root, "has"));
    }
}
```

Compressed & Radix Tree

When you are a creating a trie, similiarly to a binary tree, you face issues with height. Moving forward, you might see that you can reduce the height by compressing combinations of words.